

TASK 1: 3D CAD MODEL CREATION

Component: Bearing Housing

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1. Objective

The purpose of this task is to prepare a simple 3D CAD model of a bearing housing, define its main dimensions and select a suitable engineering material.

2. Material Selection

Selected material: Gray Cast Iron (ASTM A48 Class 30). Gray cast iron is commonly used for stationary machine housings because it has good vibration damping, good machinability and adequate strength for this type of support component.

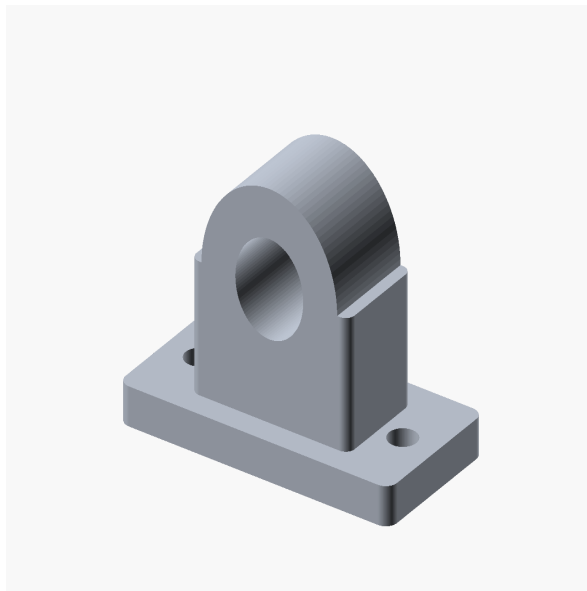


Figure 1. Isometric view of the bearing housing model.

3. Main Dimensions

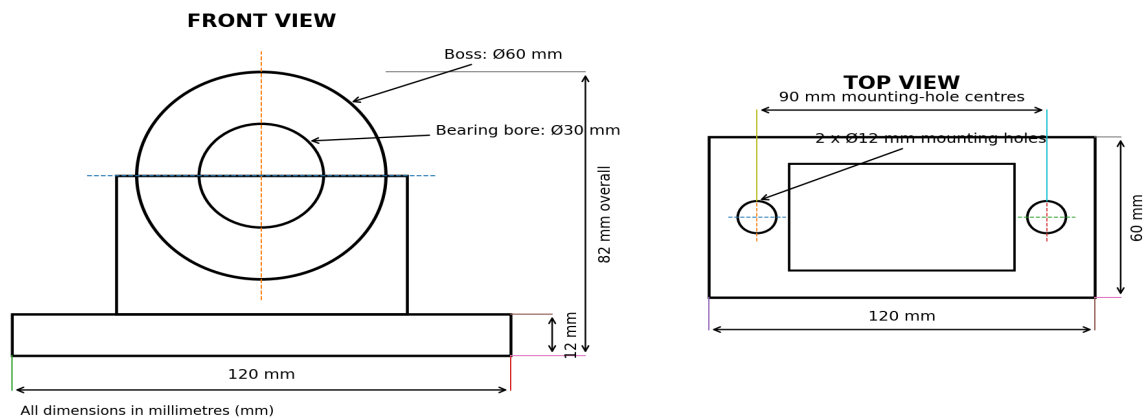
Feature	Dimension
Base plate	120 x 60 x 12 mm
Main support	70 x 40 mm
Support height above base	40 mm
Outer bearing boss	Ø60 mm
Bearing bore	Ø30 mm
Mounting holes	2 x Ø12 mm
Mounting hole centre distance	90 mm
Overall height	about 82 mm

4. Modelling Procedure

1. Create a 120 x 60 mm rectangle for the base and extrude it to 12 mm thickness.
2. Add a centered support section on the top face and extrude it upward by 40 mm.
3. Create the circular bearing boss with an outside diameter of 60 mm.
4. Cut a 30 mm diameter concentric hole through the boss for the bearing/shaft location.
5. Create two 12 mm diameter mounting holes through the base. Keep the hole centres 90 mm apart.
6. Apply small fillets to the required outside edges and check the final dimensions.
7. Assign gray cast iron as the material and save the CAD model.

5. Dimensioned Drawing

BEARING HOUSING - DIMENSIONED ORTHOGRAPHIC DRAWING



Material: Gray Cast Iron (ASTM A48 Class 30) | CAD model: STEP/STL

Figure 2. Front and top views showing the principal dimensions.

6. Conclusion

A simple bearing housing was prepared with a 30 mm bearing bore, a 60 mm outer boss and two mounting holes in the base. The geometry is suitable for demonstrating standard CAD operations such as sketching, extrusion, cutting and filleting. Gray cast iron was selected as a practical material for the housing.

CAD files included with submission: bearing_housing.step and bearing_housing.stl